

# Lecture

## The Web Server

*This content is protected and may not be shared, uploaded, or distributed.*

# Outline

- Available Web Servers
- Server Features
  - Document Root
  - Authentication
  - Proxy Servers
  - Caching
  - CGI Scripting
  - Application Program Interface
- Configuring a Server
- Analyzing a Server's Performance
- Server Log Files

# What Does the WWW Server Do?

- Enables browser requests
- Mainly provides
  - Support for retrieving hypertext documents
  - Manages access to the Web site
  - Provides several mechanisms for executing server-side scripts
    - Common Gateway Interface (CGI)
    - Application Program Interface (API)
    - Direct Module Interfaces (SAPI)
  - provides log files and usage statistics

# What to Look for in a Web Server

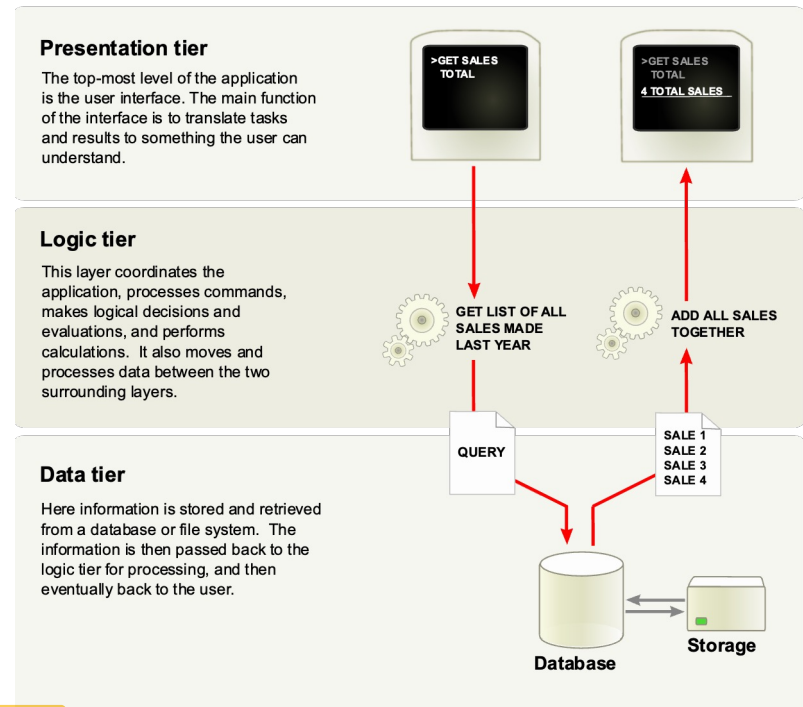
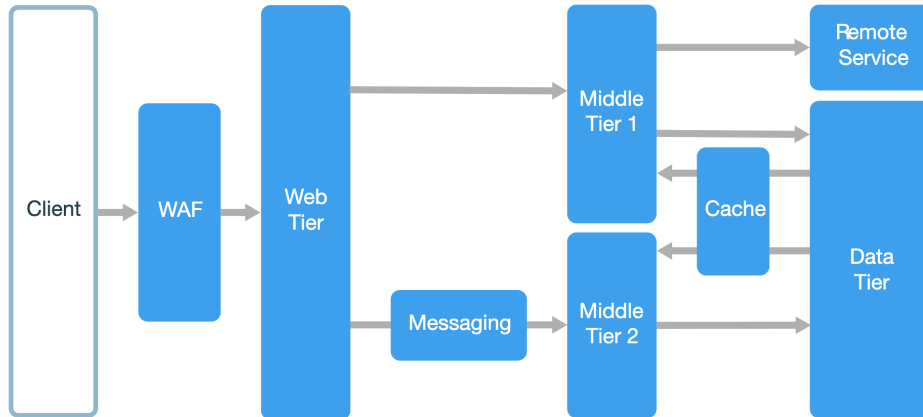
- Main features
  - platform they run on;
  - complete support for HTTP 1.1 / HTTP 2
  - Multithreading, load balancing
- Security features
  - ability to provide IP address restriction
  - ability to provide domain name restriction
  - Support for secure transactions: SSL
  - Ability to act as a proxy server

# How Servers Handle Multiple Requests

- For each request, a complete copy of the server is made and executed
  - Initially a parent process waits for requests; for each new request a child process is spawned
- Or a single server program handles many requests simultaneously (multithreaded)
  - the server must keep track of all requests and switch between them as needed
  - writing multithreaded programs is easier in a language that supports multiple threads, e.g., Java, C#, Swift, Kotlin, ...

# Application Web Server

- An **application server** is **software** that typically **interfaces** one or more **databases** to **convey processed data** to and from a **presentation layer**
- It performs **business logic**



- **Webs server** sits directly on the cloud, as "**presentation**" tier
- **Application server** (middle tier) performs business "**logic**"
- "Data" tier usually connect to a DBMS

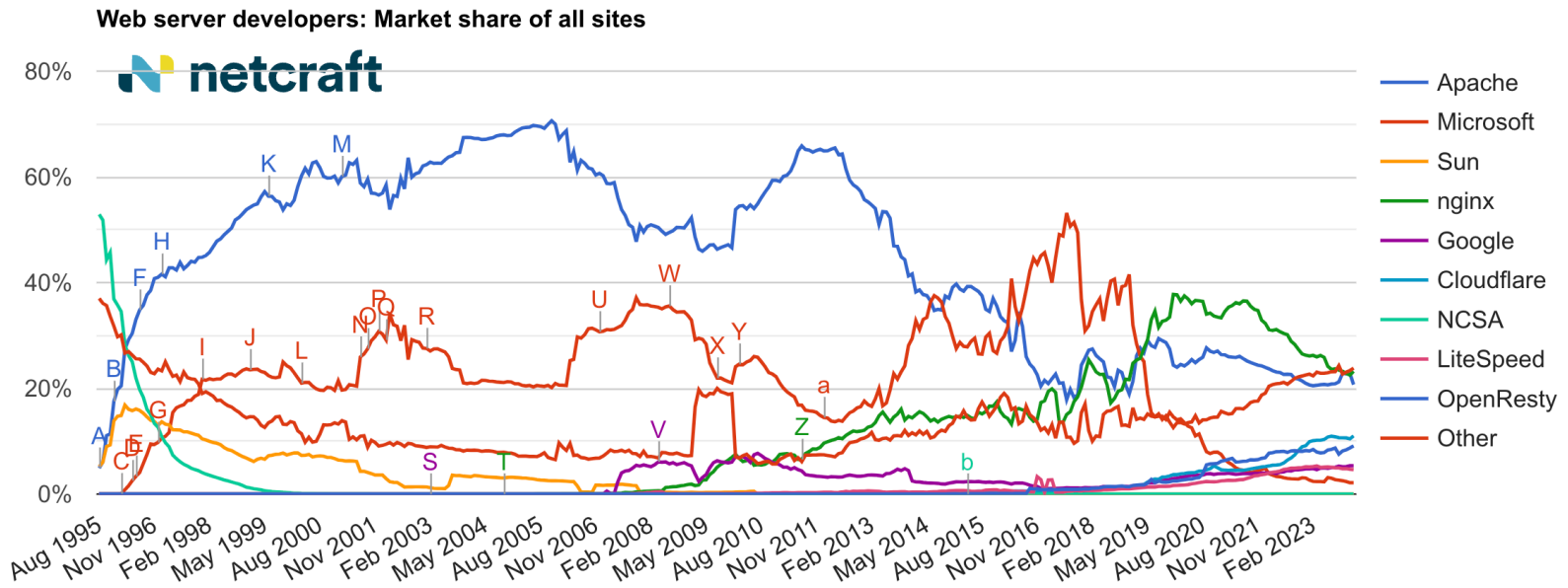
# Some Available Web & Application Servers

| Publisher                  | Product                    | Platform                  | URL                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------|----------------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Web Servers</b>         |                            |                           |                                                                                                                                                                                                                                                                                                                                                            |
| <i>apache</i>              | <b>apache</b> (2.4)        | Win32/UNIX                | <a href="http://httpd.apache.org/">http://httpd.apache.org/</a>                                                                                                                                                                                                                                                                                            |
| <i>Microsoft</i>           | <b>IIS</b> 7-10.0          | Win32                     | <a href="http://www.iis.net">http://www.iis.net</a>                                                                                                                                                                                                                                                                                                        |
| <i>NGINX</i>               | <b>nginx</b> (open source) | Linux, BSD, Win32         | <a href="http://nginx.org">http://nginx.org</a>                                                                                                                                                                                                                                                                                                            |
| <i>NGINX (F5)</i>          | <b>NGINX Plus</b>          | Ubuntu, AWS<br>GCP, Azure | <a href="http://www.nginx.com">http://www.nginx.com</a>                                                                                                                                                                                                                                                                                                    |
| <b>Application Servers</b> |                            |                           |                                                                                                                                                                                                                                                                                                                                                            |
| <i>Oracle</i>              | <b>GlassFish</b> 4.1       | Solaris/Win32/Linux       | <a href="https://javaee.github.io/glassfish/">https://javaee.github.io/glassfish/</a> (now Open-Source Jakarta EE Reference Implementation)                                                                                                                                                                                                                |
| <i>IBM</i>                 | <b>WebSphere</b>           | Win32/UNIX/Linux          | <a href="https://www.ibm.com/cloud/websphere-application-server">https://www.ibm.com/cloud/websphere-application-server</a><br><a href="https://www.ibm.com/docs/en/was-zos/9.0.5?topic=servers-websphere-application-server-community-edition">https://www.ibm.com/docs/en/was-zos/9.0.5?topic=servers-websphere-application-server-community-edition</a> |
| <i>Oracle</i>              | <b>WebLogic Server</b>     | Win32/UNIX/Linux          | <a href="https://www.oracle.com/java/weblogic/">https://www.oracle.com/java/weblogic/</a>                                                                                                                                                                                                                                                                  |

For a comparison of web servers see <http://www.serverwatch.com/>, click on Server Comparison Tool

# Web Server Usage

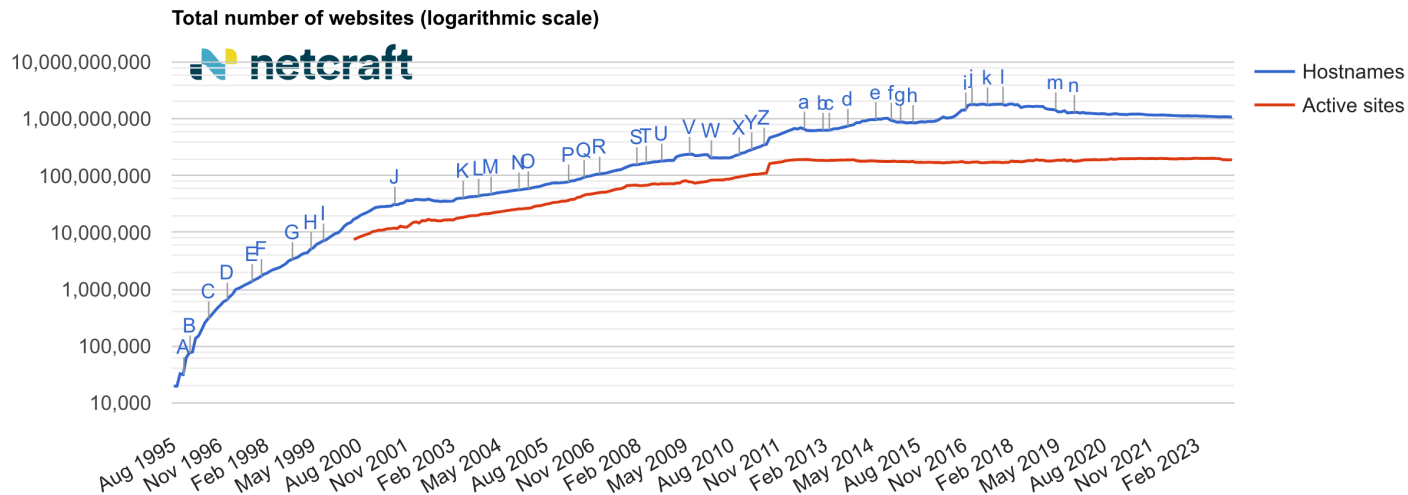
- Netcraft has identified more than 1,079,154,539 sites (1/2024)
  - Apache no longer dominates Internet servers
  - Nginx / Microsoft on top, Apache #3
  - Statistics on intranets are difficult to determine
  - <http://news.netcraft.com/archives/category/web-server-survey/>





# Web Server Survey Takeaways

- From Netcraft **January 2024** and previous surveys we can make these observations
  - **Nginx, Microsoft IIS and Apache** have each achieved their widespread popularity in the server market due to their general availability. Nginx continues to gain, while Apache continues to lose market share.
  - **Google's custom server software ("GSE")** operates on over 37 million hostnames and 2.0 million domains. Google open-sourced the **Google Servlet Engine**, at the end of 2008, but the software has not received any official public updates in 11 years.
  - **Taobao**, China's largest online marketplace and part of Alibaba Group, developed their own **fork of nginx**. The fork, known as **Tengine**, was released as open source, leading to wider adoption. Tengine runs on 56.8 million sites, making it the fourth most popular web server by this metric, despite it having only 1.2 million domains.
  - **OpenResty**, based on nginx 1.21.4, a scalable Platform by Extending NGINX with Lua.
  - **LiteSpeed** market share continues to increase, though still small.
  - **Cloudflare**, another nginx branch with extensive modifications.



# Web Server Features

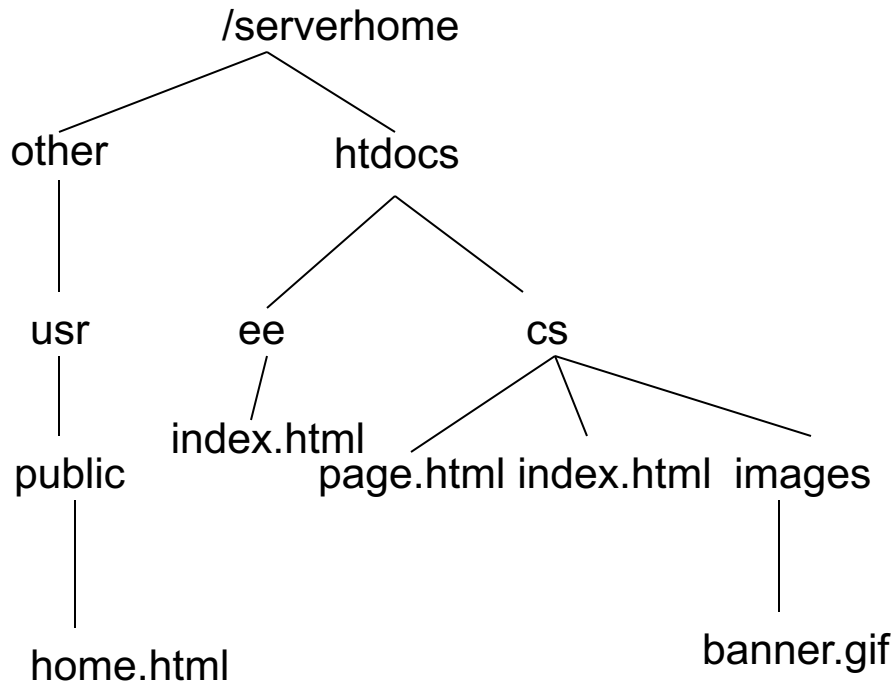
- Document Root
- Authentication
- Proxy Servers
- Caching
- CGI Scripting
- Application Program Interface

## Web Server Features - Document Root

- The Web server has access to a tree of files that it can deliver
  - the files are created by content providers
  - files may contain text, images, sound, video, other programs, etc.
- the document tree is organized by the web site administrator
- The root of the document tree is given to the web server when it starts

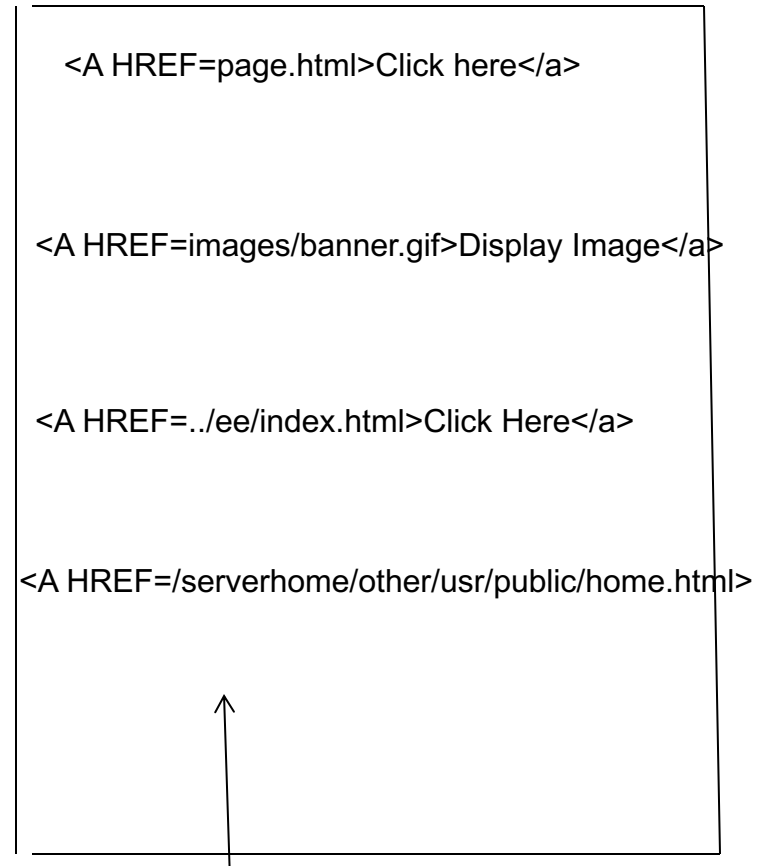
# Managing the Document Tree

`http://domain/cs` is mapped to `/serverhome/htdocs/cs/index.html`



The document root is `/serverhome/htdocs`

index.html



*The last link is an error*

# Web Server Features (1)

## Virtually Hosted Document Roots

- Hosting multiple web sites by the same web server
- It uses the host name or IP address to distinguish the document roots, e.g.

GET /index.html HTTP/1.0

**Host:** www.hardware.com

might return documents from /htdocs/hardware, while

GET /index.html HTTP/1.0

**Host:** www.antiques.com

might return documents from /htdocs/antiques

- This might be configured in, say apache by writing

```
<VirtualHost www.hardware.com>
```

```
ServerName www.hardware.com
```

```
DocumentRoot /htdocs/hardware
```

```
</VirtualHost>
```

```
<VirtualHost www.antiques.com>
```

```
ServerName www.antiques.com
```

```
DocumentRoot /htdocs/antiques
```

```
</VirtualHost>
```

## Web Server Features (2)

### Basic User Authentication

- Basic authentication is supported by all HTTP servers
  - The server administrator creates **secure directories** accessible via password files maintained by the server
  - Client requests a resource from a secure directory; e.g., **GET /secure/test.html HTTP/1.0**
  - Server responds with authentication request to the client; e.g., **HTTP/1.0 401 Unauthorized**
  - Browser requests username and password, scrambles them, and retries the request  
**GET /secure/test.html HTTP/1.0**  
**Authorization: Basic 0<V%L:EB.G-E8W) )\$J**
  - Server decodes name and password and checks it against its password file

# Web Server Features (3)

## Creating Server-Side Applications

- Web Servers offer several mechanisms
  - REST Applications
  - Application Programming Interface (API)
  - Common Gateway Interface (CGI)
  - J2EE / .NET interfaces
  - Chrome V8 JavaScript Engine (Google)
    - Node.js
  - Direct Module Interfaces (PHP)
    - Apache offers a PHP module
- Microsoft IIS server-side supports:
  - API applications compiled as DLL
  - Active Server Pages (written in VBScript)
  - ASP.NET applications (written in C++, VB.NET, C#, or HTML/CSS/JavaScript)

# Configuring a Server

- No matter which operating system or server, you will need to define the
  1. location of the server (server root)
  2. location of documents to deliver (document root)
  3. location of CGI scripts or server-side components to execute
- You may also wish to define
  - Access restrictions
  - Fancy indexing
  - Sys admin e-mail
  - Other features



# Port Numbers

- A typical computer will have thousands of ports
- System ports: 0 - 1023, are reserved for certain functions, typically assigned by IANA (Internet Assigned Number Authority), <http://www.iana.org/>
- Well-known TCP / UDP ports:

[http://en.wikipedia.org/wiki/List\\_of\\_TCP\\_and\\_UDP\\_port\\_numbers#Well-known\\_ports](http://en.wikipedia.org/wiki/List_of_TCP_and_UDP_port_numbers#Well-known_ports)

- programs running on these ports must be started under **root**

- TCP/UDP ports used by Apple products:

<http://support.apple.com/kb/HT6175>

- Dynamic ports **49152 - 65535**, ports that are freely available (check OS X possible conflicts)

## Some Well-Known Port Numbers

| Port | Protocol  | Purpose                          |
|------|-----------|----------------------------------|
| 20   | ftp       | ftp data connection              |
| 21   | ftp       | ftp control connection           |
| 23   | telnet    | telnet session                   |
| 25   | SMTP      | simple mail transfer protocol    |
| 53   | DNS       | domain name service              |
| 70   | gopher    | gopher protocol                  |
| 79   | finger    | finger protocol                  |
| 80   | http      | hypertext transfer protocol      |
| 88   | kerberos  | kerberos authentication protocol |
| 110  | pop3      | post office protocol             |
| 113  | ident     | remote identity service          |
| 119  | nntp      | network news transfer protocol   |
| 143  | IMAP      | email access                     |
| 161  | snmp      | simple network management proto  |
| 162  | snmp      | snmp traps                       |
| 194  | irc       | internet relay chat              |
| 280  | http mgmt | http management                  |
| 389  | ldap      | lightweight directory access pr  |

## Standard Port Numbers (cont'd)

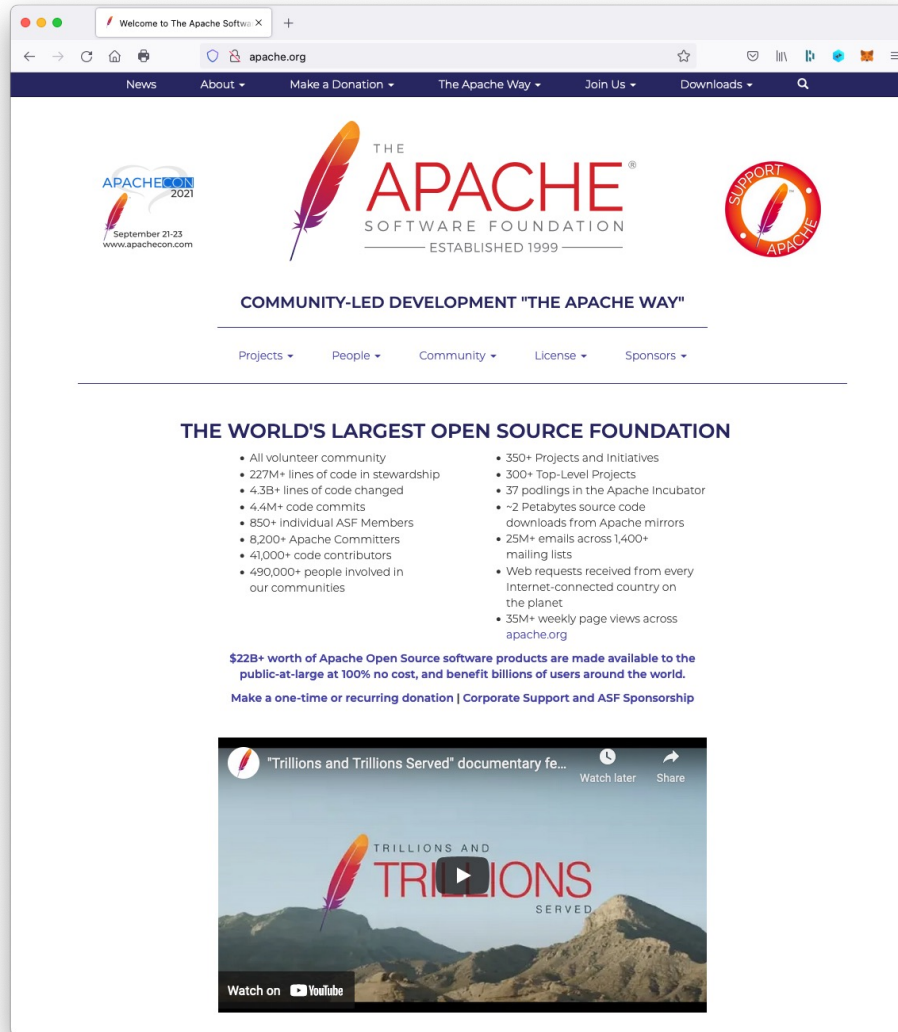
| Port | Protocol       | Purpose                       |
|------|----------------|-------------------------------|
| 427  | srvloc         | server location protocol      |
| 443  | https          | secure http (SSL)             |
| 465  | smtps          | secure email (SSL)            |
| 535  | iiop           | internet inter-orb protocol   |
| 551  | cybercash      | secure money transaction      |
| 563  | snews          | secure news (SSL)             |
| 614  | ssl shell      | ssl shell                     |
| 636  | ldaps          | secure ldap (SSL)             |
| 989  | ftps           | secure ftp data connection    |
| 990  | ftps           | secure ftp control connection |
| 992  | telnets secure | telnet (SSL)                  |
| 993  | imaps          | secure imap (SSL)             |
| 994  | ircs           | secure irc (SSL)              |
| 995  | pop3s          | secure pop3 (ssl)             |
| 1080 | socks          | socks protocol, circuit level |

# **The Apache Web Server**

# Apache Web Server

- Apache is based on code and ideas developed at NCSA in `httpd` version 1.3 (early 1995)
- The name “Apache” comes from the fact that it is A set of **PAtCHES** to **NCSA's httpd**
- Apache is free and is downloadable from <http://httpd.apache.org>
- Apache included in every version of macOS:  
<https://jasonmccreary.me/articles/install-apache-php-mysql-mac-os-x-mojave/>  
<https://discussions.apple.com/docs/DOC-250004361>
- Using Apache in Windows 10/11  
<https://www.thewindowsclub.com/install-apache-on-windows-10>
- **Apache HTTP Server version 2.4.58 is current.**  
Version **2.4.43** or newer is required in order to operate a **TLS 1.3** web server with **OpenSSL 1.1.1**.

# Apache Home Page



Available for  
many  
platforms:  
Windows,  
Solaris,  
Linux,  
macOS,  
etc.

# Apache **httpd** server

- The Apache httpd server
  - is a powerful, flexible, HTTP **1.1 & HTTP/2** compliant web server
  - implements **HTTP/2**, but **not HTTP/3**
  - is highly configurable and extensible with third-party modules
  - can be customized by writing 'modules' using the Apache module API
  - provides full source code and comes with an unrestrictive license
  - runs on Windows, Netware, Mac OS X, Linux, and most versions of Unix, as well as several other operating systems

# Configuring Apache

- server functionality is available through modules which are either built-into or loaded into the server
- server instructions
  - apache reads its run-time configuration instructions from text files
- no GUI available (visual start/stop available on Windows)
- 182 configuration directives in the basic package
- On UNIX, Apache starts several processes to implement the server and handle requests
- On Windows there are only two processes, a parent and a child that handles all requests. Within the child all requests are handled by separate “threads”
- See <http://httpd.apache.org/docs/2.4/configuring.html>



# How does Apache Work?

- Apache runs under a multitasking operating system, e.g., UNIX, macOS, Windows
- the binary is called httpd
- Key directories include:
  - conf contains configuration files
  - htdocs contains html files
  - logs contains logging data
  - cgi-bin contains executable scripts
  - icons set of \*.gif files used e.g. in directory listings
  - src contains actual source code if you download the source distribution
- Apache idles, listening to an IP address and port

# Configuring Apache - conf Directory

- In the conf directory there is a file provided with model configuration parameters

- httpd.conf

- This file must be edited

- Partial contents of the mime.types file

|                 |              |
|-----------------|--------------|
| text/html       | html htm     |
| text/plain      | txt          |
| text/richtext   | rtx          |
| video/mpeg      | mpeg mpg mpe |
| video/quicktime | qt mov       |
| video/x-msvideo | avi          |
| x-world/x-vrml  | wrl vrml     |

# Apache Settings - httpd.conf

- Directives are keywords followed by their value, e.g.

```
Listen 9637
```

```
ServerAdmin horowitz@usc.edu
```

```
ServerRoot /home/scf-03/csci571/WebServer/apache_2.2
```

```
ErrorLog logs/error_log
```

```
DocumentRoot "/home/scf-17/csci571c/apache2/htdocs"
```

```
ScriptAlias /cgi-bin/ "/home/scf-17/csci571c/apache2/cgi-bin/"
```

- the above are ones that *must* be set

## Some Apache Settings - httpd.conf

# **Timeout**: The number of seconds before receives and sends time out

Timeout 300

# **KeepAlive**: Whether or not to allow persistent connections (more than one request per connection). Set to "Off" to deactivate.

KeepAlive On

# **MaxKeepAliveRequests**: The maximum number of requests to allow during a persistent connection. Set to 0 to allow an **unlimited** amount. They recomend you leave this number **high**, for maximum performance.

MaxKeepAliveRequests 100

# **KeepAliveTimeout**: Number of seconds to wait for the next request

KeepAliveTimeout 15

## Apache Settings - httpd.conf

# **Server-pool size regulation**. Apache dynamically adapts to the load it sees --- it tries to maintain enough server processes to handle the current load, plus a few spare servers to handle transient load spikes. It does this by periodically checking how many servers are waiting for a request. If there are **fewer** than **MinSpareServers**, it **creates** a new spare. If there are **more** than **MaxSpareServers**, some of the spares **die** off.

```
MinSpareServers 2
```

```
MaxSpareServers 2
```

```
StartServers    2 #number of servers to start
```

```
MaxClients      2
```

# Apache Settings - httpd.conf

# **DocumentRoot**: The directory out of which you will serve your **documents**.

**DocumentRoot** /home/cscixxx/WebServer/apache/htdocs

# **UserDir**: The name of the directory which is appended onto a user's home directory if a **~user** request is received.

**UserDir** public\_html

# **DirectoryIndex**: Name of the file or files to use as a pre-written HTML

**DirectoryIndex** index.html

# **FancyIndexing** is whether you want fancy directory indexing or standard

**IndexOptions** FancyIndexing

# Apache Settings - Scripting

```
# ScriptAlias: This controls which directories
  contain server scripts. Format: ScriptAlias
  fakeName realName
ScriptAlias /cgi-bin/
  /home/cscixxx/WebServer/apache/cgi-bin/
# If you want to use server side includes, or CGI
  outside ScriptAliased directories, uncomment the
  following lines.
# AddType allows you to tweak mime.types without
  actually editing it, or to make certain files to be
  certain types. Format: AddType type/subtype ext1
# To use CGI scripts:
#AddHandler cgi-script .cgi
# To use server-parsed HTML files
#AddType text/html .shtml
#AddHandler server-parsed .shtml
```

# Authentication in Apache

- There are two methods for controlling access to directories
  1. the file `access.conf` in the `conf/` directory can be used
    - placing access `directives` in a `<Directory>` block is the preferred method
  2. per-directory access rules can be set by a `file` placed in a `specific directory`
    - the name of the file is set by the directive `AccessFileName`
    - `.htaccess` is the default name
    - however, using `.htaccess` slows down the server;
    - NOTE: initially `.htaccess` is turned OFF
- When the server attempts to retrieve a document, it looks for an access control file in the `directory` or the `parent directory`.
- The file it looks for is set by the directive `AccessFileName`



## Some Sectioning Directives

- The server configuration files use `sectioning directives` to define `special access rights` within a `section or sub-directory`
- Example: `<Directory>` specifies directory to which directives apply
- Options controls which server features are available
  - `None`
  - `All`
  - `FollowSymLinks` follow symbolic links
  - `ExecCGI` execution of CGI scripts allowed
  - `Includes` server-side includes are enabled
  - `Indexes` will automatically return a list of files in the directory

## Limit Sectioning Directive

- `<Limit>` controls which clients can access a directory; directives within `LIMIT` include:
  - order in which `deny` and `allow` are evaluated
  - `deny from host1, host2, ...`
  - `allow from host1, host2, ...`
  - `require named-users or group-users or AuthUserFile`
  - `referer` allows access only from this directory
  - `satisfy all or any`

## Using LIMIT

- Host Filtering is used to limit document trees to certain machines
- Example 1: to limit access to the cscixxx public\_html documents to USC only

```
<Directory /usr/~cscixxx/public_html/>
```

```
<Limit GET>
```

```
order deny,allow
```

```
deny from all
```

```
allow from .usc.edu
```

```
</Limit>
```

```
</Directory>
```

- If someone tries to access documents in this directory from outside of usc they get a 403 Forbidden message

## Using LIMIT (cont'd)

- Example 2: to limit documents so USC people CANNOT access them

```
<Directory /usr/~cscixxx/public_html/>  
<Limit GET>  
order allow,deny  
allow from all  
deny from from .usc.edu  
</Limit>  
</Directory>
```

## Using LIMIT (cont'd)

- Example 3: a directive used to limit access to only USC and ISI domains

```
<limit GET>
```

```
order deny, allow
```

```
deny from all
```

```
allow from 128.125
```

```
allow from 128.9
```

```
</Limit>
```

## Using LIMIT (cont'd)

- Example 4: Suppose you want to restrict files in a directory called secure/ to the user named student1 and password XXXYYY
- Step 1. create a file called .htaccess in directory secure/ that contains these 7 lines:

```
AuthUserFile /otherdir/.htpasswd      must be a full
    UNIX pathname
AuthGroupFile /dev/null                there is no group file
AuthName ByPassword                   any name is OK here
AuthType Basic                        Basic or Digest is OK
<Limit GET>
require user student1
</Limit>
```

## Using LIMIT (cont'd)

- Step 2. Next create the password file `/otherdir/.htpasswd` using the `htpasswd` program distributed with apache.
- Type  
`htpasswd -c /otherdir/.htpasswd student1`  
and enter the password twice.
- You are done.
- Step 3. To add other names and passwords run `htpasswd` without the `-c` option.

## Using LIMIT (cont'd)

- To generalize this example to include several students,
- Step 4. create a group file called /otherdir/.htgroup which looks like:

```
my-users: student1 student2 student3 student4
```

Alter the .htaccess file to look like

```
AuthUserFile /otherdir/.htpasswd
```

```
AuthGroupFile /otherdir/.htgroup
```

Now it points  
to the group file

```
AuthName ByPassword
```

```
AuthType Basic
```

```
<Limit GET>
```

```
require group my-users
```

```
</Limit>
```



## Contents of .htaccess

- The .htaccess file contains pointers to the users file, the group file, and defines the name and type of authorization

```
nunki.usc.edu(10): more .htaccess
```

```
AuthUserFile /home-scf-03/cscixxx/pwd/.htpasswd
```

```
AuthGroupFile /home-scf-03/cscixxx/pwd/.htgroup
```

```
AuthName OnlineCourse
```

```
AuthType Basic
```

## Contents of .htpasswd

- This file contains a set of names and passwords

nunki.usc.edu(13): more .htpasswd

admin:kjpokufbLdrXU

chungw:2Ds5QooEqT1HM

csci351:ykGZC.bMvV8SY

csci351notes:7814Vi6oJtq9c

csci565:t.OSxBunNhigg

csci571:NUKyNCCLvgLH.

nunki.usc.edu(13): more .htgroup

admin: admin csci571 csci565

student: admin csci571 csci565

# Adding a Smart Error Page

- When someone requests a page or directory that does not exist it is better to return an error page than a 404 Not Found message
- For an example try: `http://www.usc.edu/x.html`
- Look at `/conf/httpd.conf`
- find the section that describes the properties of your domain, e.g.

```
<VirtualHost www.test.com>
ServerAdmin webmaster@test.com
DocumentRoot /web/test
ServerName www.test.com
ErrorLog /log/test/error_log
transferLog /log/test/access_log
AgentLog /log/test/agent_log
RefererLog /log/test/refer_log
</VirtualHost>
```

## Adding a Smart Error Page

- Insert the line

```
ErrorDocument 404 /error-page.html
```

- place the error page at

```
DocumentRoot/ErrorDocument or in this case  
/web/test/error-page.html
```

Sample contents of error-page.html:

```
<HTML><HEAD><TITLE>Test Error Page</TITLE></HEAD>  
<BODY>Error. You have requested a page that  
    cannot be found. You may wish to return to the  
    <A HREF=http://www.test.com/>home page by  
    clicking here.</A>  
</BODY></HTML>
```

# Apache Approaches to Server-Side Processing

- **CGI** - the standard way to invoke an external process, e.g., perl & perl.exe
- **Fast CGI, Java Servlets** - Apache passes a request to another running server which executes the request and returns the result
  - see mod\_fastcgi at <http://www.fastcgi.com>
  - For Jakarta Servlets see Tomcat at <https://tomcat.apache.org/>
- **Interpreters** embedded in the server itself. There are roughly two categories
  - Modules that answer or modify requests directly, e.g., mod\_perl (Perl)
  - Modules aimed to process commands embedded in HTML pages before serving it to the client. e.g., mod\_php (PHP)

# Virtual Hosting

- *Virtual hosting* is a way of setting up your server so that it can appear to be multiple Web sites at once, depending upon how it's accessed by clients
- ISPs do this a lot
- The Apache Web server software allows you to create as many such virtual site identities as your system's limits will allow;
- Example of a virtual host definition.

```
<VirtualHost 10.0.0.1>
```

```
    ServerName WWW.Mydomain.Com
```

```
    ServerAdmin Webmaster@Mydomain.Com
```

```
    DocumentRoot /home/httpd/mydomain
```

```
    CustomLog logs/mydomain_log CLF
```

```
</VirtualHost>
```

# Virtual Hosting (cont'd)

- there are two ways to do virtual hosting
  - *address-based* virtual hosting uses IP addresses
  - *name-based* hosting allow a single IP address to have multiple identities, e.g.

```
NameVirtualHost 10.0.0.1
```

```
<VirtualHost 10.0.0.1>
```

```
    ServerName WWW.Mydomain.Com
```

```
    ServerAlias WWW.X.Com WWW.Y.Org
```

```
    ServerAdmin Webmaster@Mydomain.Com
```

```
    DocumentRoot /home/httpd/mydomain
```

```
    CustomLog logs/mydomain_log_CLF
```

```
</VirtualHost>
```

```
<VirtualHost 10.0.0.1>
```

```
    ServerName WWW.Other.Org
```

```
    ServerAlias WWW.Z.Net WWW.A.Org
```

```
    ServerAdmin Webmaster@Other.Org
```

```
    DocumentRoot /home/httpd/other
```

```
    CustomLog logs/other_log_CLF
```

```
</VirtualHost>
```

this example assigns

two domain names to the

same IP address

# Content Negotiation

- The client provides a set of preferences concerning the preferred attributes of a document, and the server tries to come up with the best match from the versions available
  - If there's only one version of a document, then that's what the server is going to send
- The set of all versions of a document that are available to the server is called the resource's *list of variants*.
- The ways in which they differ, such as by language or encoding, are called its *dimensions*.
- The Apache server can negotiate along the
  - content-type,
  - encoding, and
  - language dimensions



# Content Negotiation

- The primary use of negotiation on the Web today is to select translations of text.
  - Typically, a work is prepared in a default language such as English, and the result put in a file named `text.html` or some such. Then translated versions are prepared, and each is put into an appropriate-named file: `text.html.da` for Danish, `text.html.fr` for French, `text.html.en-gb` for English (Great Britain dialect), and so on.
- You can either prepare a `list of variants` yourself, or you can allow Apache's `mod_negotiation` module to `calculate the list itself by scanning the directory`.
- The latter method is more `common`, since it is easier and `automatically` pick up `new variants` as soon as they are available, but it *does* impose a slight `performance penalty due to the directory scan`.

# Apache as a Proxy Server

- proxy servers are frequently used in fire walled environments to provide a **single point of contact** between the **users** inside and the **Web** outside the firewall.
  - The users ask for a page, the browser sends the request to the proxy server, the proxy server sends the request to the origin server, and the data flows back the same way.
- A second function commonly associated with proxy servers is **caching**.
- To use the proxy functions in your Apache Web server, the **mod\_proxy** module must be activated, either by being statically linked in (less common than formerly), or turned on at **run-time** with directives, e.g.

**AddModule mod\_proxy.c**

# Apache as a Proxy Server, Proxy Directives

- **ProxyRequests** - This is the 'master directive,' which enables and disables proxy operation overall. It takes a single keyword argument, which must be either On or Off
- **ProxyRemote** - tells the Apache server that certain requests are to be forwarded to yet another system rather than being handled locally, allowing for cascading of proxies
- **ProxyBlock** - allows you to selectively block the fetching of certain resources through the proxy. It takes a list of space-separated case-insensitive phrases or URI substrings which should be blocked.
  - "ProxyBlock foo" would block all requests in which "foo" appeared, in any combination of upper- and lower-case, including "http://foo.com/bar/" and "http://bar.org/zardofoo.html". You can block all proxy requests with "ProxyBlock \*"

# Controlling the Apache Cache

- **CacheRoot** – Defines the directory that's the top of the cache hierarchy
  - Apache uses its own naming conventions for the stuff it stores.
- **CacheSize** – Specifies the largest number of kilobytes of space you want to allocate to the proxy cache.
- **CacheDirLevels** – Controls how deep the cache directory tree is allowed to grow
- **CacheDirLength** – Used to define the number of characters used in naming each level of cache subdirectory
- **NoCache** – Identifies proxy-accessed documents that shouldn't be cached. The format is the same as for the ProxyBlock directive.

## Proxy Host Example

- Usually, a proxy server is set up as a completely **separate virtual host** from the rest of the server's operation (which is likely to be handling normal requests).
- You can do this with a `<VirtualHost>` stanza. The `<IfModule>` container will only be evaluated if the proxy module is part of the running server

```
<IfModule mod_proxy.c>
```

```
    Listen 10.0.0.1:8080
```

```
    <VirtualHost 10.0.0.1:8080>
```

```
        ProxyRequests On
```

```
        DocumentRoot /home/httpd/htdocs/proxy
```

```
    </VirtualHost>
```

```
</IfModule>
```

# Analyzing Web Server Log Files

- A sample entry from a log file

| Host           | date/time                        | method | file | HTTP ver  | status | bytes |
|----------------|----------------------------------|--------|------|-----------|--------|-------|
| marmot.usc.edu | - - [17/Feb/2009:23:21:09 -0800] | "GET   | /    | HTTP/1.0" | 200    | 1316  |

- There are different log file formats
  - W3C Extended Log File Format
    - Date/time, server port, URL and query, protocol version, user agent and cookie, referrer, etc.
  - Microsoft IIS log file format
    - IP, user, date/time, server name, server IP, bytes received, bytes sent, protocol, method, etc.
  - Other file formats
    - Cerfnet, apache extended, CERN common, Lotus Domino, Netscape Common, etc.

## Definition of Terms

- *A hit*: a request for a page, image, cgi script, etc.
- *A page view*: a request for an html page
- Unique host vs. unique visitor: aol.com is a unique host, so this statistic may not be useful; cookies are best way to track unique visitors
- *An impression*: is a view of an advertisement
- *A click-through* is when an advertisement is actually clicked on

## Common Log Format

- Each visit to your site produces a record of the request in log file
- log files in Apache are typically stored in the logs directory
- Some fields include:
  - Host                      client hostname or IP address
  - ident        remote identity, often a dash
  - usr                      authenticated username
  - time        date/time request is received
  - req                      HTTP request
  - s1                      server's HTTP response status code
  - c1                      Content-length
- Example entries from access log

```
marmot.usc.edu - - [17/Feb/2008:23:21:09 -0800]  
"GET / HTTP/1.0" 200 1316
```

```
128.125.229.195 - - [23/Feb/2008:10:32:09 -0800]  
"GET / HTTP/1.1" 200 1316
```



## Other Entries in access Log

- ciscoserv-229-180.usc.edu - - [17/Feb/2008:23:16:51 -0800] "GET /cgi-bin/secure/admin/student.pl HTTP/1.1" 401 362
- 128.125.229.195 - admin [23/Feb/2008:10:56:21 -0800] "POST /cgi-bin/secure/admin/student.pl HTTP/1.1" 200 807
- 209.85.18.67 - - [27/Feb/2008:01:12:38 -0800] "GET /ftp/templates.zip HTTP/1.1" 200 30780
- nunki.usc.edu - - [28/Feb/2008:00:48:22 -0800] "GET / HTTP/1.0" 200 955
- nunki.usc.edu - - [28/Feb/2008:00:48:26 -0800] "GET /cgi-bin/Count.cgi?df=count.dat HTTP/1.0" 200 207

## access\_log Contents

- Here are some lines from a single user session

```
lv1-sun702.usc.edu - - [09/Feb/2008:19:15:36
-0800] "GET /newsdigest/index.html HTTP/1.0"
200 3630

lv1-sun702.usc.edu - - [09/Feb/2008:19:15:37
-0800] "GET / HTTP/1.0" 200 955

lv1-sun702.usc.edu - - [09/Feb/2008:19:15:39
-0800] "GET /newsdigest/index.html HTTP/1.0"
200 3630
```
- A *user session* includes all hits and requests made by a user terminated by “n” minutes of inactivity

# Sample error\_log Contents

- Here are some lines from the error\_log file

- **-The file was not found**

- ```
[Mon Jan 29 11:47:36 2008] httpd: access to /home/scf-13/csci665/www/pleader/horohome.html failed for internet-gw.watson.ibm.com, reason: No file matching URL: /pleader/horohome.html from -
```

- **authorization entered is incorrect**

- ```
[Thu Sep 24 10:17:41 2008] [error] [client 128.125.50.47] user csci571 not found: /cgi-bin/secure/test-cgi
```

- **Perl script fails**

- ```
[Sun Sep 30 01:15:01 2008] [error] [client 71.140.65.229] Premature end of script headers: /auto/home-scf-22/csci571/WebServer/Apache_1.3.36/cgi-bin/cgipm/test6.pl
```

## More Examples of Error Log File Entries

- Undefined subroutine &main::Export\_to\_Excel called at /home/scf-03/csci571/WebServer/apache\_1.2.5/cgi-bin/admin/student\_db.pl line 75.
- [Tue Dec 29 02:07:47 2008] access to /auto/home-scf-03/csci571/WebServer/apache\_1.2.5/htdocs/robots.txt failed for 193.189.227.21, reason: File does not exist
- [Tue Dec 29 20:11:22 2008] send directory lost connection to client 128.125.223.237

# Performance and Security Considerations

- Don't log more than needed
- Look for 401 (unauthorized) and 403 forbidden status codes
- Look at the response times
  - Which parts of the web site have the longest response times?
  - Where are the bottlenecks
- Look for failed page hits (404, 5xx)
- Look for cached hits
- Look for browsers and platforms of clients

# Tools to Analyze Log Files

- Some commercially available tools for IIS
  - **WebTrends**  
<http://www.webtrends.com>
  - **Dynatrace**  
<https://www.dynatrace.com/>

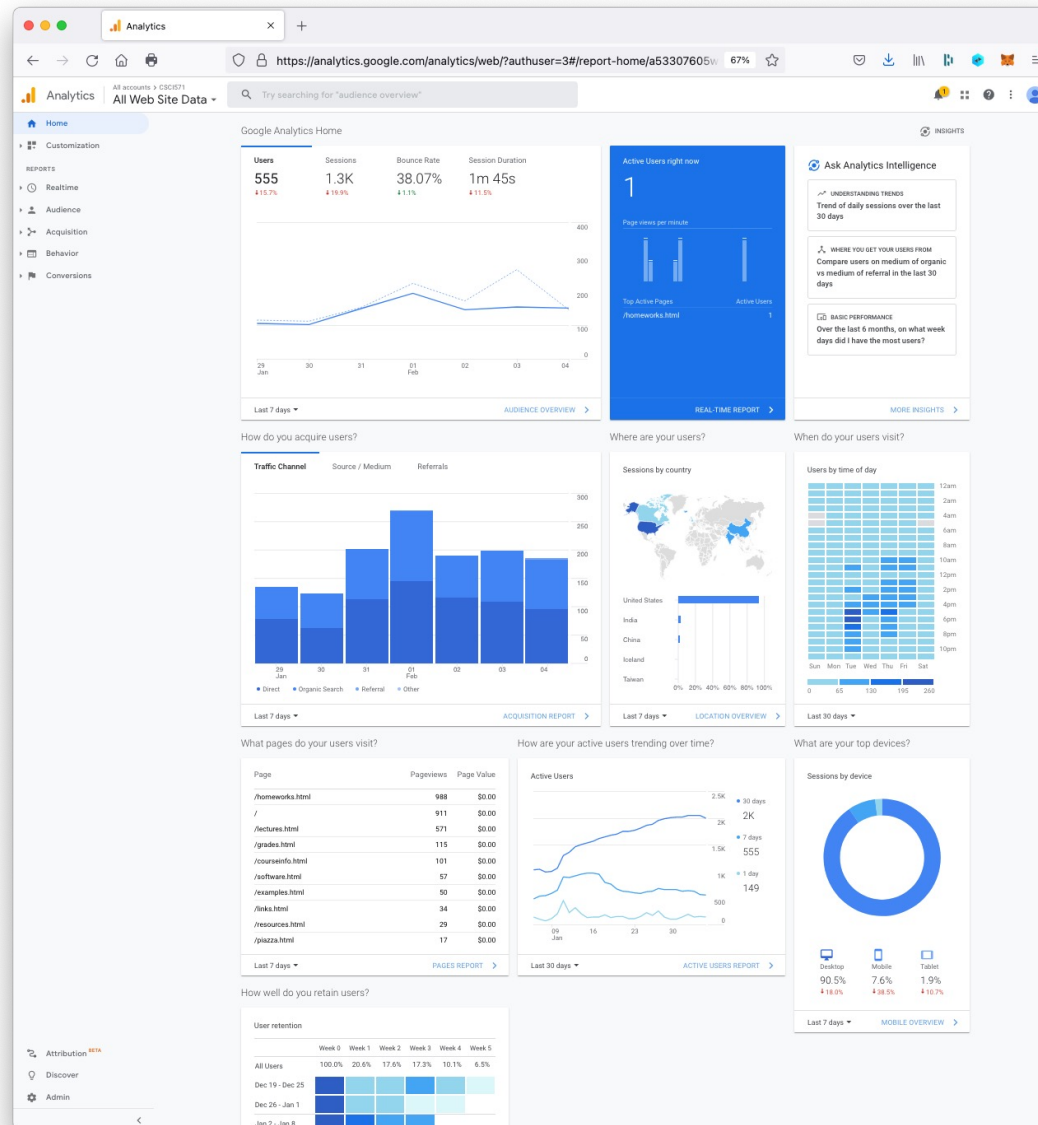
# Google Analytics

- For any web server
  - **Google Analytics**. See:  
<http://www.google.com/analytics/>
- Install **global site tag**, after <head> tag, on every site page.  
<https://developers.google.com/tag-platform/gtagjs>

```
<!-- Global site tag (gtag.js) - Google Analytics -->
<script async src="https://www.googletagmanager.com/gtag/js?id=GA_MEASUREMENT_ID"></script>
<script>
  window.dataLayer = window.dataLayer || [];
  function gtag(){window.dataLayer.push(arguments);}
  gtag('js', new Date());

  gtag('config', 'GA_MEASUREMENT_ID');
</script>
```

# Google Analytics Dashboard





# Google Analytics Behavior Flow

